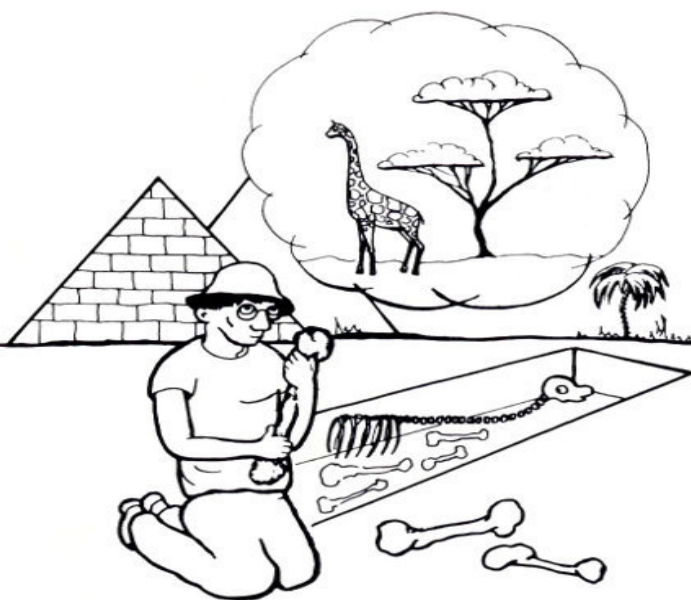


HOW OLD IS IT?

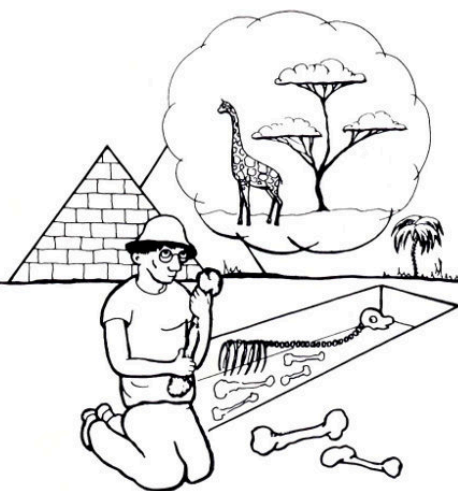
THE STORY OF DATING IN ARCHAEOLOGY



MUSEUM OF NEW MEXICO PRESS
POPULAR SERIES PAMPHLET NO. 2

HOW OLD IS IT?

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HOW OLD IS IT?

THE STORY OF DATING IN ARCHAEOLOGY

**MUSEUM OF NEW MEXICO PRESS
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HOW OLD IS IT? DATING IN ARCHAEOLOGY *by James Schoenwetter*

There is a whole field of science devoted to the invention and development of dating methods—or “clocks” as we may think of them. It is called *geochronology*, the science of dating events. There are relatively few geochronologists, scientists trained in the use of all kinds of dating methods and in the theories upon which these methods are based.

Geochronologists tell us that there are two major types of clocks: those that tick at an absolute rate of speed which can be measured, and those which tick only once in a while. A clock of the first type yields what is called an *absolute* date, revealing the number of hours, days, years, centuries or millennia since an event occurred. A clock of the second type yields what is called a *relative* date, placing an event as before or after another event, but does not tell us exactly how far they are apart in time nor how long ago they occurred.

Depending upon how accurate his date must be to solve the problem he has set for himself, the archaeologist will select absolute or relative dating methods. Often, of course, the type of clock he wishes to use is not available, and he must use the next best type. Probably he will try to use a number of clocks of different kinds on the problem since each clock will act as a check on the others.

The absolute clock utilized most widely in archaeology is the historical record. Men have used calendars for a long time, and have often left records with written dates. On tombstones at a site in old Virginia, on the pedestals of statues and other monuments from classical Greece and Rome, on the walls of the tombs of Egyptian kings, dates are clearly inscribed which can be related to the sites dug into by the archaeologists. These dates must often be recalculated in terms of the Christian calendar which we use. Most calendars in use in the Mediterranean, the Near East and China during classical antiquity have been successfully correlated with the one we

use today, and a date inscribed or noted on such sites can be considered in our own terms. Other calendars, such as those developed in the ancient cultures of the Maya on the Yucatan Peninsula, have yet to be accurately correlated with our own. Such calendars can be used on their own terms of course, and a site which has an inscription in the Mayan calendar is known to be so many years older or younger than another one with a different date in that calendar. We speak of such a situation as a *floating chronology*. That is, the sequence of events and the number of years which separate them are known, but the dates of those events in absolute time are unknown.



Tree rings afford another kind of absolute clock, the *dendrological* 3 method. Each year a tree adds a growth ring. Depending on the amount of water the tree has available to it for cell growth, the ring will be wider or narrower. Certain trees whose water requirements are high live near streams or other places where their roots can tap a constant supply of water. Such trees, referred to as *complacent*, have annual rings which are all of about the same width. Other *sensitive* trees live in places where they must depend almost wholly on rainfall for their water supply, as on the slopes of hills or in the clefts of rocks. Such trees have annual rings which vary in width depending upon the amount of rain they receive. In any given area, especially in arid and semiarid regions, some years have more rainfall than others. The sensitive trees will produce wider rings during years when there is more rainfall and narrower rings in years when there is less. Often there are periods of a decade or so when all of the sensitive trees will produce the same pattern of ring growth; for example, three years of narrow rings, one year of wide, two more of narrow and three more of wide. Such a pattern is called a *signature*.

Signatures are the basis of tree ring chronologies. All the trees in a region did not begin growing at the same time of course, but every time there is a series of years which will produce a signature, all the sensitive trees still alive will have that signature. Let us say we cut down a sensitive tree in 1960 and, by counting back the rings, find signatures at 1940-45, 1910-14, 1880-89, 1821-27, 1795-1800 and 1750-58. Next we recover a beam from an abandoned Spanish Mission built in 1810. It happens to be from a sensitive tree, and we can spot the 1795-1800 and the 1750-58 signatures near the outer rings. Now we have two records which can be said to be *crossdated*. Let us assume that the Spanish Mission log give us signatures as far back as 1350. An abandoned Indian pueblo produces a log with signatures crossdating those of the Spanish Mission log and continuing the record back to 1250. Older and older archeological sites will yield older and older signatures with each log crossdating some of the signatures of younger logs.

At present we have a tree ring calendar for certain species of trees extending back to about 100 B.C. This is called a *master tree ring chronology*. A log from an archaeological site may contain only one or two

signatures and be, in itself, a floating chronology, but by comparison with the master chronology one can determine the cutting date. Having a series of cutting dates for the construction timbers in an archaeological site yields the probable dates at which the site was built and occupied.

4

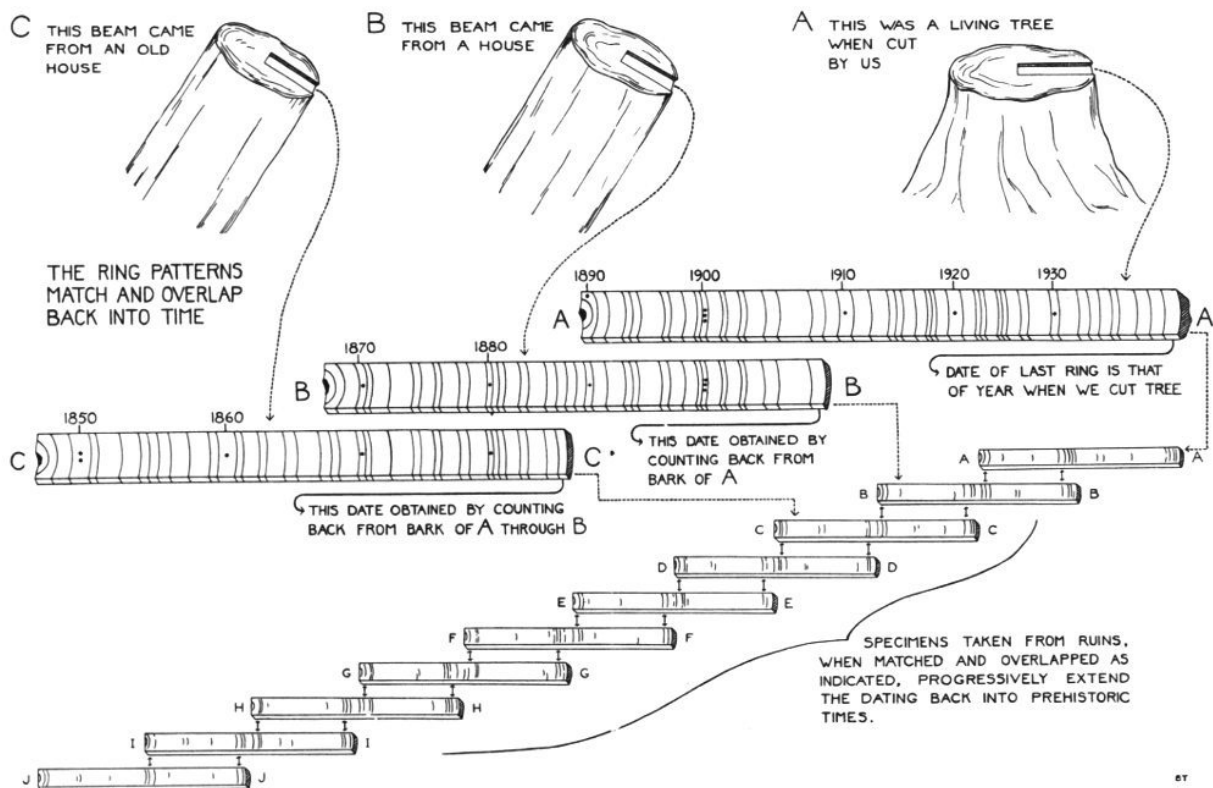


Diagram to Illustrate Tree Ring Dating. Reproduced by special permission from Stallings, "Dating Prehistoric Ruins by Tree-Rings."

C THIS BEAM CAME FROM AN OLD HOUSE

B THIS BEAM CAME FROM A HOUSE

A THIS WAS A LIVING TREE WHEN CUT BY US

THE RING PATTERNS MATCH AND OVERLAP BACK INTO TIME

SPECIMENS TAKEN FROM RUINS, WHEN MATCHED AND OVERLAPPED AS INDICATED, PROGRESSIVELY EXTEND THE DATING BACK INTO PREHISTORIC TIMES.

Not all kinds of trees can be used for dendrochronology. Pine, fir and pinyon are the most useful; juniper can sometimes be used. Oak, cottonwood, willow and others are very difficult to date and frequently cannot be used at all. 5

Another widely used absolute clock is that based upon the orderly decomposition of carbon. This is the *radiocarbon* or C-14 method of dating. Molecules of various substances are made up of atoms. We know now that not all atoms of a substance are precisely the same. We speak of *isotopes* of an atom. To clarify this let us think of the atoms of carbon as being made up of a mass of ping-pong balls. Some atoms will have more ping-pong balls than others, but all will have enough and in the proper order to be carbon atoms. Each of the atoms with 14 ping-pong balls we shall refer to as the carbon-14 isotope of carbon. There will be other isotopes of carbon atoms too.

The carbon-14 isotope is about average in life span. It has been determined that in any group of C-14 isotopes half of them will lose two of their ping-pong balls and become C-12 isotopes in 5,730 years, plus or minus 40. This is known as the *half-life* of the C-14 isotope. The “plus or minus 40” allows for laboratory error in terms of years.

Now all living things contain carbon atoms, and some of those atoms are C-14 isotopes. The amount of C-14 isotopes in a living organism quickly reaches a stable percentage after which there is no increase or decrease while the organism is alive. After it dies, the C-14 supply is not replenished, and with the passing of 5,730 plus or minus 40 years, it has half the number of C-14 isotopes it had when alive. In 11,460 plus or minus 80 years, it will have one quarter as many as it had when alive.

The geochronologist takes a certain weight of carbon-bearing matter from an organism which once lived. With simple chemistry he can determine the number of carbon atoms in the material, usually charcoal, wood, bone or shell. He places the material in a chamber equipped with geiger counters and records the number of C-14 isotopes converting to C-12 isotopes within a certain number of hours or days. Since he knows how many carbon atoms there are in the specimen, he knows how many C-14 isotopes there would be if the specimen were alive. He also knows that as the number 6

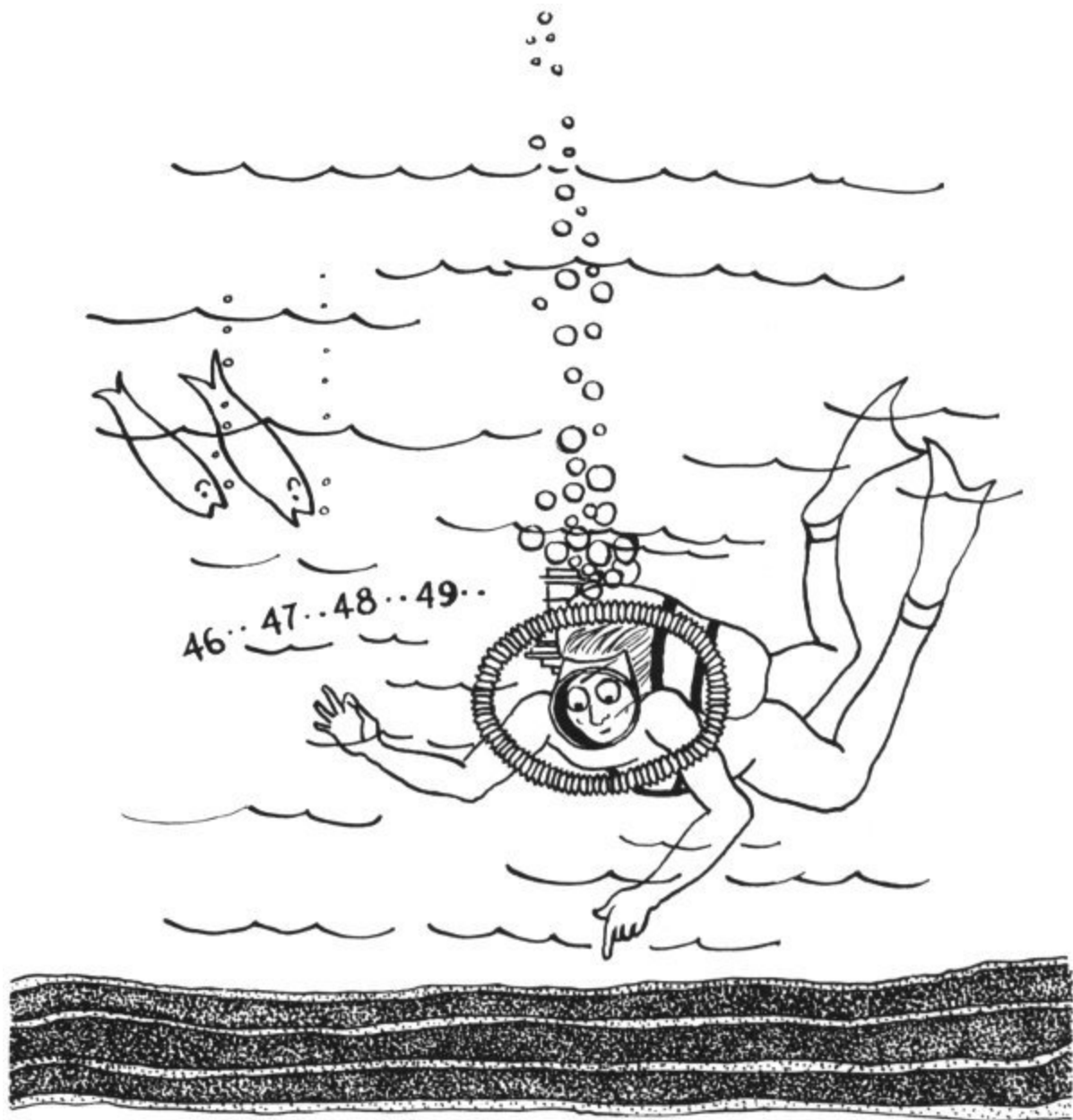
of C-14 atoms decreases the number of clicks on the geiger counter will decrease too. For example, if there are 2,000 C-14 isotopes, the decomposition of half of these over a period of approximately 5,730 years would register 1,000 clicks on the counter. In the next 5,730 year period there would be 1,000 isotopes left, and only 500 of them would decompose to register as geiger counter clicks.

The geochronologist does the counting and analysis of the results and sends the information back to the archaeologist in the form of the number of years that have elapsed since the carbon was part of a living creature; for example, 1,500 plus or minus 150 years BP (before present). The archaeologist, converting this to the Christian calendar in 1964, would come up with A.D. 464 plus or minus 150 years. When was the sample actually alive? We don't know exactly, but statistically we have a ninety-five percent chance of being right if we say sometime between A.D. 164 and 764. This clock ticks in centuries. But the radiocarbon clock doesn't tick very long, even in centuries, before running down. By the time 30,000 to 40,000 years have gone by, the C-14 in any sample is almost gone, and there is too little left to give enough geiger counter clicks unless one is willing to wait a lifetime to record two or three clicks.

The C-14 dating technique measures time by radioactive decomposition of materials. There are other clocks which depend on the chemical decomposition of materials. The forgery of the Piltdown Man fossil was detected by a dating method which depends on the decomposition of bone protein and its replacement by fluorine. Fluorine is an element which occurs naturally as a gas, but which combines readily with other elements to form compounds. Some of these elements are common in bones. Since fluorine is one of nature's most reactive elements, it tends to escape from the compound it is in and to form other compounds. As the protein in a bone decays, it is often replaced by fluorine. The amount of fluorine in old bones, then, is expected to be more than in young bones since it has had more time to accumulate. Since fluorine does not accumulate at a constant rate, it affords only a relative measure of age.

In the case of Piltdown, a group of bones was discovered at a site, and was said to contain those of one individual who lived about 60,000

years ago. Almost 100 years after they were discovered, these bones were put to the fluorine test. It was found that (1) some of the bones had less fluorine than others, so not all were of the same antiquity and could not have belonged to the same individual, (2) the younger bones had as much fluorine in them as modern bones, and (3) the older bones had more fluorine in them than bones known to be 60,000 years old.



Another form of absolute dating of importance to archaeology is that called the *varve* method. This can only be utilized as an absolute clock under very special circumstances however. Varves are like tree rings in a way. In a lake which is sufficiently deep, or at the edge of a glacier, particles of sediment are being deposited continuously as a sort of fallout from the water. 8 During the winter, when the glacier freezes or the density of the water in the lake increases because of the cooler temperature, less particles are deposited, and those which are deposited are usually of a characteristic color or texture. During the summer, when the glacier melts or the lake warms up, more and different particles are deposited. The bands of deposited sediment are called varves; every year two varves are formed. Starting from the top, one can count back the number of years in a varve series. If the top varve is of known date such as the present year, one has a calendar with each varve having a known date. Attempts are made to correlate one varve series with another in order to recover even longer series. If the archaeologist is lucky, and it is not rare in Europe, there will be materials from a site buried in the local varve sequence. Counting back gives an absolute age for the artifacts embedded in the site and thus an approximate age for the site. The European varve chronology is believed to extend back to about 9650 B.C.

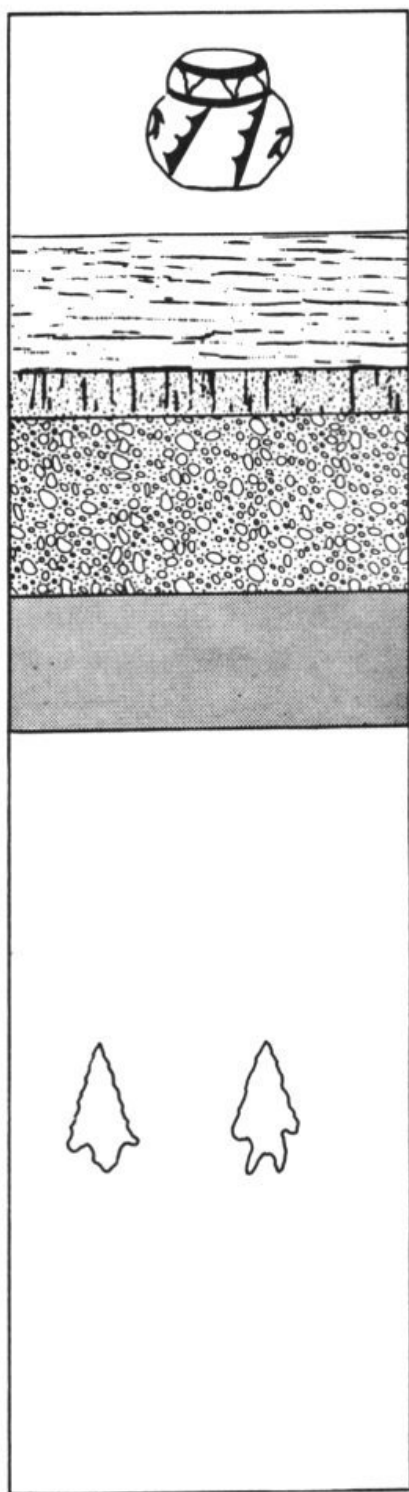
Most of the clocks which the archaeologist uses to produce relative dates, the before-or-after kind, have as their theoretical basis the principle of *stratigraphy*. In effect the principle of stratigraphy assumes two things: that the rocks of the earth are constantly wearing down by erosion, and that things which appear to be alike actually are alike and are probably more or less the same age.

If rocks are constantly wearing down, it follows that the surface of the ground is constantly building up. Thus the surface we walk on is a younger, higher surface than that which our ancestors walked on. When we dig below the surface, those things we find which are at higher levels are younger than those which we find at lower levels. The deeper we dig, the older things get.

There is no reason to believe that the rate of deposition on the surface is the same everywhere. If we dig two feet in one place we may be at a level

which is now five feet below the surface in another location. If we find a particular object, say a type of pottery, on the surface at site A and the same kind of pottery five feet below the surface at site B, we can use the second of our assumptions and maintain that both pieces are of the same age. Then any objects found at higher levels than five feet at site B are younger than the piece of pottery and are younger than anything found at site A. This is the principle of stratigraphy.

Like tree rings, objects in *stratigraphic sequence* can be crossdated. 9 These sequences may be of various kinds as any object will do. Distinctive bands of sediment, distinctive artifacts, types of fossils, specific details of chemistry or any other phenomenon may be used with varying amounts of success. Suppose we have the following sequences of objects at sites A and B:



A

Black earth

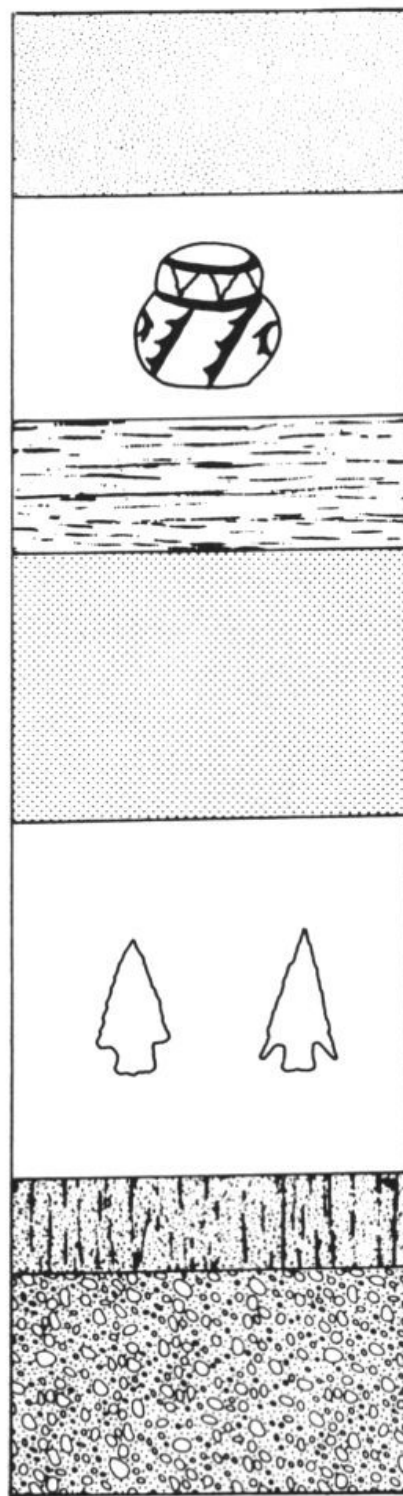
Caliche

Eroded layer

Cobbles

Soil

Brown silt



B

Brown dust

Black earth

Caliche

Yellow silt

Brown silt

Eroded layer

Cobbles

A

B

<i>Black earth</i>	<i>Brown dust</i>
<i>Caliche</i>	<i>Black earth</i>
<i>Eroded layer</i>	<i>Caliche</i>
<i>Cobbles</i>	<i>Yellow silt</i>
<i>Soil</i>	<i>Brown silt</i>
<i>Brown silt</i>	<i>Eroded layer</i>
	<i>Cobbles</i>

10

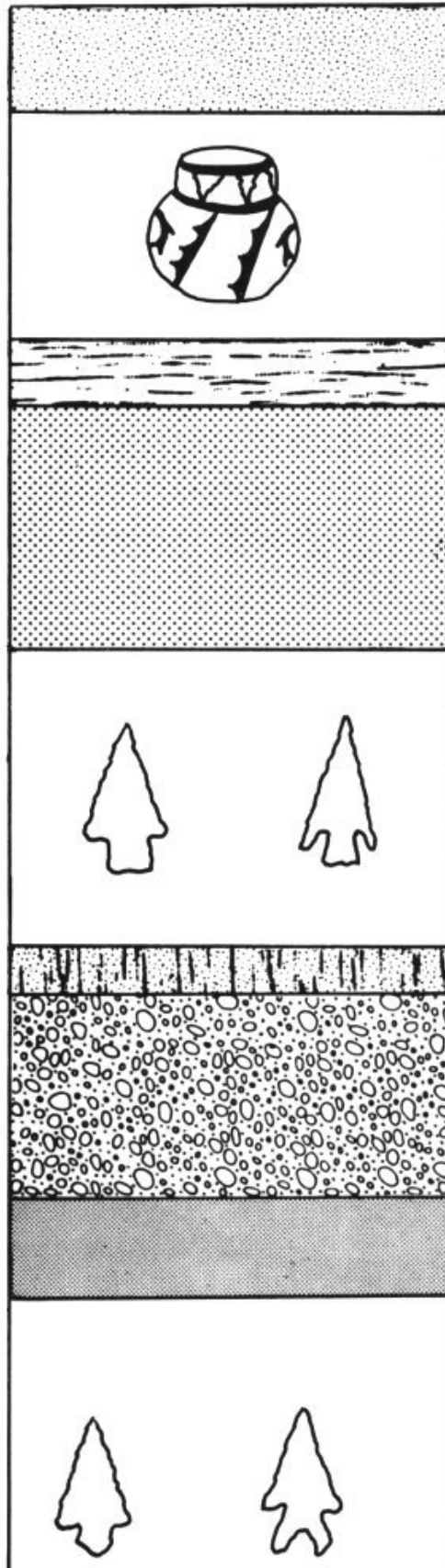
Now there are some things that are similar about these two *profiles* and other things that are different. Both profiles contain layers of black earth, caliche, an eroded layer and cobbles. Both profiles contain pottery, and both contain arrow points. The type of pottery in profile A is the same as that in profile B, but in B there is brown dust above the black earth. In profile B the brown silt is above the eroded layer, while in A it is below the eroded layer. In profile A the arrow points have different shapes than the ones in profile B. What we need to correlate the sequences are *horizon markers*, objects that are enough alike to be in the same time range.

The pots in both cases look the same and are embedded in the same kind of sediment, the black earth. They form one horizon. The caliche may be a horizon marker, but this is not positive since at profile A it is above the eroded layer, while at profile B it is above the yellow silt. The eroded layer sits above cobbles at both profiles, which makes the eroded layer-cobbles complex a pretty good horizon marker. The brown silt is not a horizon marker because it is above the eroded layer at one profile and below it at the other. Furthermore, the arrow points which it contains are not all of the same kind. The correlated sequence, using the horizon markers, must be as shown in the accompanying [chart](#).

The stratigraphy proves that the arrowheads in the upper brown silt must be younger than those in the lower brown silt. The next time we find arrowheads of the types recovered in the upper brown silt we will know, regardless of the stratigraphic sequence in the new locality, that they must be younger than the types found in the lower brown silt, and they must also be older than pottery of the type found in the black earth. We won't know

how many years old they are, but we have dated them in the sense that we know they are older than some things and younger than others.

The principle of stratigraphy is one of the archaeologist's most useful theoretical tools as it allows one site to be compared with others. Our example has shown how artifacts (pots and arrowheads in this case) and sediment types can be placed in stratigraphic order to provide a clock for archaeological dating. Other objects which are commonly used as horizon markers in stratigraphic sequences are fossils. Remains of extinct animals are often used to prove extreme antiquity. Remains of plants and animals will often indicate similarity of ecological conditions at two sites and allow them to be crossdated.



Brown dust

Black earth

Caliche

Yellow silt

Brown silt

Eroded layer

Cobbles

Soil

Brown silt

Brown dust
Black earth
Caliche
Yellow silt
Brown silt
Eroded layer
Cobbles
Soil
Brown silt

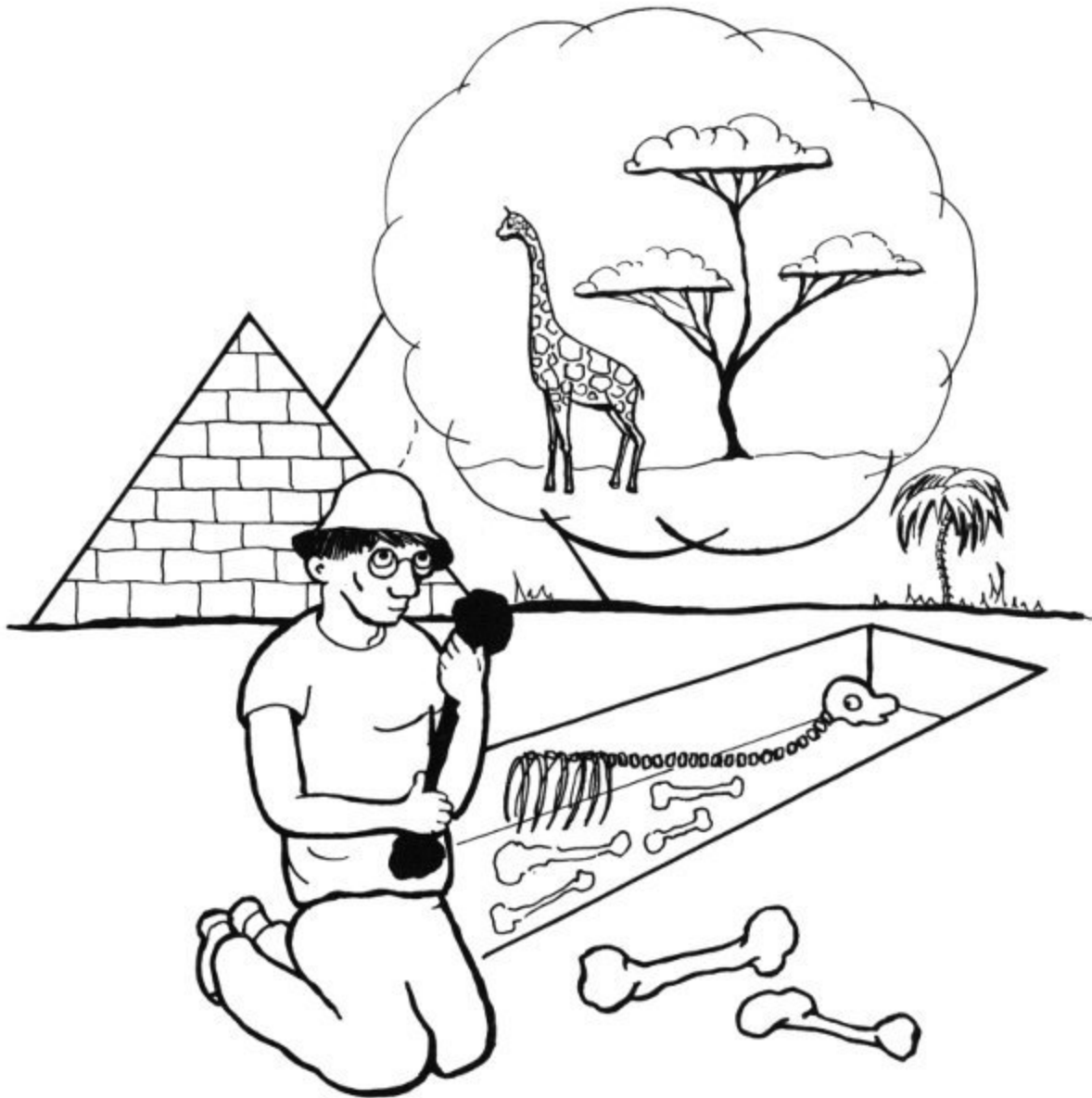
The *geologic-climatic* method of dating combines the principle of stratigraphy with an interpretation of the meaning of natural occurrences. When geologists demonstrated that the earth had recently gone through a period when enormous glaciers advanced and retreated across the northern and southern hemispheres, they began to speculate on the effects of such conditions on the landscape. The surface would be scraped down to bare rock by the advancing ice sheet. When the ice retreated, its load of rocks and cobbles would tend to be left behind, while the smaller particles of dirt would be flushed away in the rivers formed by the melting ice. If the archaeologist found tools among the boulders and cobbles, those tools should date to the period when the cobble and boulder stratum was formed, the period when the glacier was retreating. Here the geology gave clues to the climate, and if the age of the climatic event was known, artifacts associated with the geology could be dated. 12

It is through the geologic-climatic method of dating that archaeologists discovered that human beings lived in southern Europe at the margin of the last great glacier, and therefore the date of their occupation was on the order of 20,000 years ago.

It is not necessary to use the geologic aspect of geologic-climatic dating if one has other clues to the climate. Plant and animal fossils are clues to ancient climates, since living organisms have a tendency to live in climates to which they are best adapted. If we find the fossil bones of a giraffe or an ostrich in the Sahara Desert, we can conclude that at some time in the past the Sahara was not a desert but a veldt, since giraffes and ostriches live in the veldt today. If stratigraphy allows us to relate artifacts to those fossils,

we can maintain that the makers of those artifacts lived at the time the veldt existed in what is now the Sahara area. Now if we can relate the fossils to a date when such a climate could have obtained in the area, we have a date for the artifacts.

Animal fossils are relatively rare, and fossil remains of plants which can be seen with the naked eye are even rarer. But microscopic pollen grains are not particularly rare in the stratigraphic sequences of sediments, and are directly associated with artifacts in many kinds of sediment. The pollen of most plants is protected by a tough coat like that on many seeds. Millions of pollen grains are produced by local vegetation each year, and a 13 percentage of them are buried in the yearly accumulation of sediment. Once they are buried, the tough outer covering is preserved.



When pollen grains are extracted from their sediment matrix, the different types of plants which grew in the area at the time the sediment was laid down can be recognized from the distinctive characteristics of their pollen grains. The *pollen analyst* has the job of interpreting this information to discover the nature of the vegetation patterns in the past and the climates associated with them. Through dating those climates, the artifacts with which the pollen sample was associated are dated.

Some archaeological sites do not allow dating by any of these methods. The farmer plowing a field may pick up an arrowhead and wonder how old it is. Can the archaeologist date the arrowhead? The answer is a qualified 14 yes. What the archaeologist will try to do is give an educated guess as to the age of the arrowhead. He might, for example, know that arrowheads of that type are found at a site which is dated about A.D. 1000 by various methods. By correlation, he could apply that date.

If the arrowhead is not of a type which has been reliably dated, the archaeologist may rely upon the method of dating known as *seriation*. This method depends on the second assumption basic to the principle of stratigraphy, that similar objects tend to be about the same age. The farmer's arrowhead may not be exactly like any which has ever been discovered, but it probably will be more like some known ones than others. Observing the general style of the arrowhead and the way in which it is made, the archaeologist can make a pretty good estimate of when it was made.

Let us assume that the archaeologist goes into an unexplored area where no absolute or relative dating techniques are available. There he finds a number of sites with potsherds of types new to archaeology. Can he date these sites? Again he uses seriation, but this time he inverts the logical proposition. If things which look alike tend to be of the same age, things which do not look alike should tend to be of different ages.

His first task would be to separate all the different types of potsherds. Taking a specimen of each type, he lays them out in a row. If there is any difference in the sites through time, the styles of pottery will change correspondingly. But there will be some styles which change slowly and some which will influence others. For example, if we were to seriate the style of automobile rear fenders for the period 1956 to 1964, we would observe that they began to sprout taller and taller fins; then the size of the fins became reduced more and more. Some of the style aspects of pre-1956 rear fenders went along with the development of fins and others dropped out. Some of the style aspects which were developed with the fins were retained after the fins decreased in size.

Seriation of the design styles on the potsherds will result in a series of developments in style which are probably in chronological order. Broad straight line designs may give way to mixed broad and narrow straight lines, then to narrow straight lines, then to narrow wavy lines. The archaeologist could then maintain that sites which have potsherds with broad straight lines are separate in time from those with sherds having narrow wavy lines. He won't know which is the older, because the sequence could work either way. Stratigraphy or the cultural similarity between other artifacts at the sites will probably resolve this problem. 15

The reader will now have realized that many of the clocks used by archaeologists are interrelated. Relative dating clocks such as stratigraphy, pollen dating and geologic-climatic dating are utilized together where possible, and all are dependent upon the principle of stratigraphy. Crossdating is thus of vital importance and is constantly undertaken. The archaeologist tries to employ both absolute and relative clocks to find out the age of a site. Stratigraphy yields a series of relative dates for artifacts within the site; geologic-climatic and pollen dating yield a series of dates for types of sediment and samples collected in association with those artifacts; tree ring dates yield the absolute age of the site which allows the pollen, stratigraphic and geologic-climatic dates to be comprehended in terms of absolute age. Radiocarbon dates act as a check on the tree ring dates and, if they agree, lend support to the pollen and geologic-climatic dates. The pollen and geologic-climatic dates from the site are compared with similar dates from other sites as additional checks. Since the clocks used by the archaeologist tick at different rates of speed, and since not all of them are dating the same thing, the archaeologist usually ends with a series of dates for any given site.

New geochronological techniques are always being invented and perfected. Here is a list of some that are expected to become available in the next few years:

1. The *obsidian hydration* method. Obsidian, a naturally formed glass, is so constituted chemically that it takes chemicals from its environment at a slow rate. As it does so, the outside layer changes from transparent to translucent or opaque. The depth of the opaque layer, it is hoped, can be

used as a measure of the amount of time since the surface was exposed. As many artifacts were made by chipping obsidian to form a sharp edge, this method may reveal the time since an obsidian artifact was made.

2. The *thermoluminescence* method. The chemical properties of certain minerals change when the minerals are heated to high temperatures. If they are heated again they will glow, and the amount of time they glow upon reheating depends on the amount of time that has elapsed since they were heated originally. As yet the rate has not been accurately calculated, nor is it yet understood what effect different kinds of soils and atmospheres 16 may have on this rate. If the method is proven, it will be invaluable to archaeology since it will afford a way of dating pottery directly on an absolute time scale.

3. The *paleomagnetism* method. When an object containing particles which can be magnetized is heated, the magnetic particles line up according to the earth's magnetic field. When the object cools, the particles are trapped in this position. We know that the earth's magnetic field is changeable, and that at different times it has been oriented in different directions. Work is now in progress to determine how the field has varied through time and how successfully one can date materials such as pottery or hearths which were exposed to high temperatures in the past by the difference between the present magnetic field and that trapped in the object.

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Santa Fe 1965

*Illustrations by Mary Spencer
and Phyllis Hughes*

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THE STORY OF DATING IN ARCHEAOLGY ***

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